Dependability Evaluation of a Critical System through a SAN

Assigned Scenario: (R_1, O_1, W_1)

Università degli Studi di Firenze

Software: Science and Technology

Architecture, Model and Analysis of CPS (AMA-CPS) – Modulo II



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1. System Description and Formal Specifications

The system under analysis presents a mixed series-parallel topology, where the logical integrity of the path between the source s and the destination d defines the operational status. The architecture is dominated by Unit A, which acts as a Single Point of Failure (SPOF): being arranged in series with respect to the rest of the graph, its failure interrupts any information or energy flow, nullifying the redundancy offered by the parallel branches downstream.

The redundancy cluster is structured on two levels of heterogeneous parallelism:

- **Subsystem B**: Composed of two units ($\mathrm{B}_1, \mathrm{B}_2$) with intermediate failure rates ($\lambda/2$).
- **Subsystem C**: Composed of three units ($\mathrm{C}_1, \mathrm{C}_2, \mathrm{C}_3$) characterized by the highest failure frequency (λ).

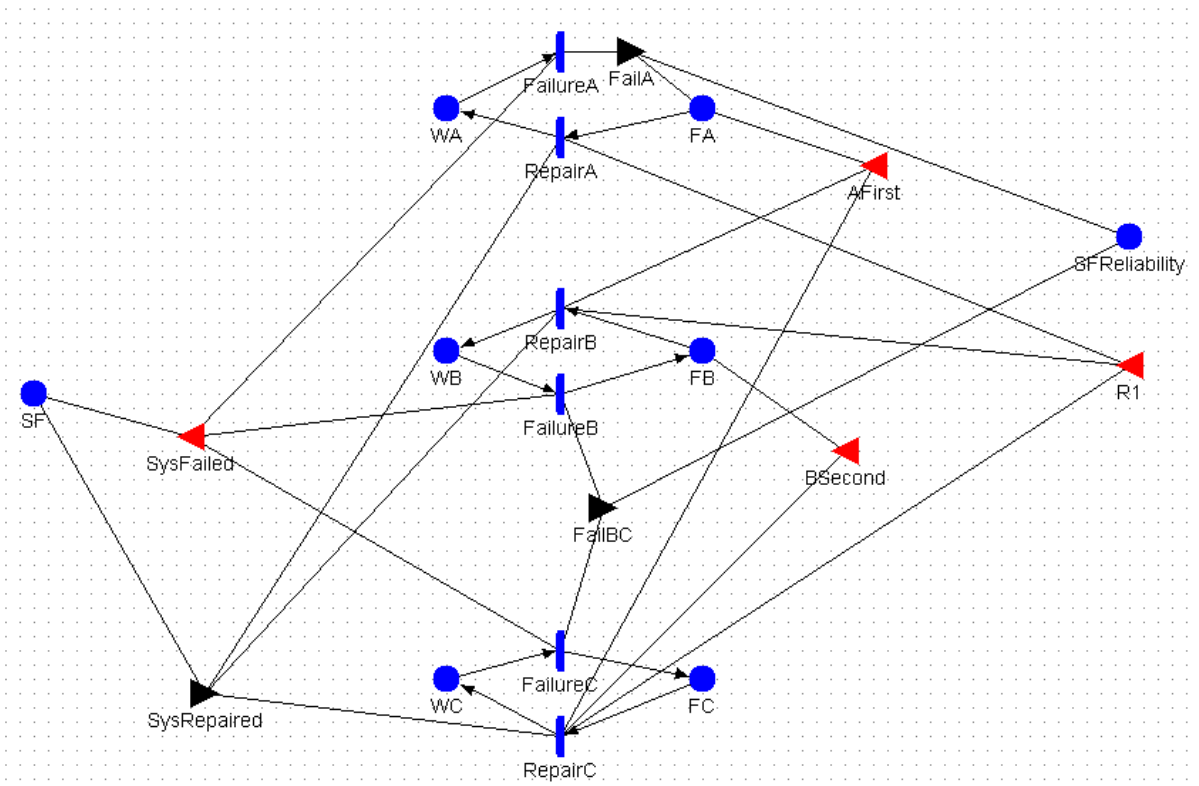
From a stochastic point of view, the model assumes constant failure and repair rates, typical of the Continuous-Time Markov Chains (CTMC) process generated starting from the SAN model.

2. Definition of the Assigned Scenario

The R1-O1-W1 configuration defines an operational model where system survival is managed through reactive and sequential intervention policies.

- Repair Trigger (Option R1): The system adopts a run-to-failure policy. Units, although they can fail individually, are not subject to repair as long as at least one active path persists between s and d. This means that the repair process is activated exclusively when the system enters a state of total failure. From a modeling perspective, this translates into a forced delay in activating repair transitions, increasing the probability of accumulating latent faults in subsystems B and C.
- Scheduling Policy (Option O1): Once the restoration phase is activated, repairs do not occur in parallel but follow a strict, predefined sequential order. The repair "server" follows the hierarchy: $A \to B_1 \to B_2 \to C_1 \to C_2 \to C_3$. This choice introduces a temporal dependence constraint: even if unit A were intact, the repairer could not intervene on units C without having first "cleared" the queue of units A and B.
- Reactivation Logic (Option W1): The system adopts a "greedy" or minimum latency recovery strategy. It does not wait for the restoration of full nominal integrity; the system returns to operational status as soon as a single functional path is re-established between the source s and the destination d. This reduces immediate downtime but exposes the system to rapid new failures, as it might restart in a vulnerable state (e.g., with unit A repaired but all branches B and C still failed).

3. SAN Model



The assigned policies R1-O1-W1 are reflected in the predicates and in the input and output gate functions, which can be inspected in the project zip folder within the SAN model file.

4. Evaluation Metrics and Experimental Design

The objective of the analysis is to evaluate the system's behavior under systematic variations of the failure (λ) and repair (μ) rates. The experimental plan foresees 9 configurations obtained by combining three levels for each parameter: $10^{-2}, 10^{-3}, 10^{-4}$ h^{-1}. To quantify the performance, three key metrics have been selected, described below with the relative observations on the obtained results.

4.1 Stationary Availability

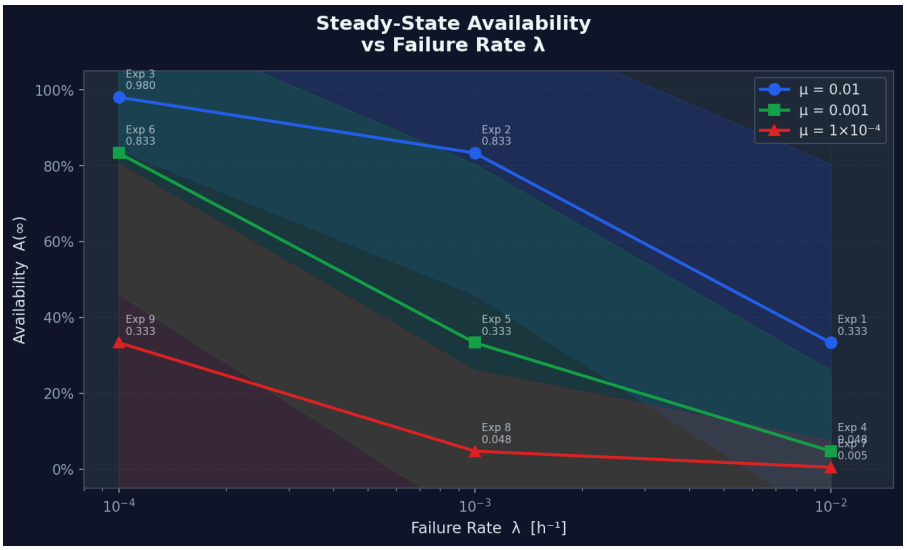
Steady-state Availability $A(\infty)$ represents the probability that the system is operational under steady-state conditions at a time instant sufficiently far from $t = 0$.

$$A(\infty) = \frac{\mu}{\lambda + \mu} = \frac{1}{1 + \rho}$$

$$\rho = \frac{\lambda}{\mu}$$

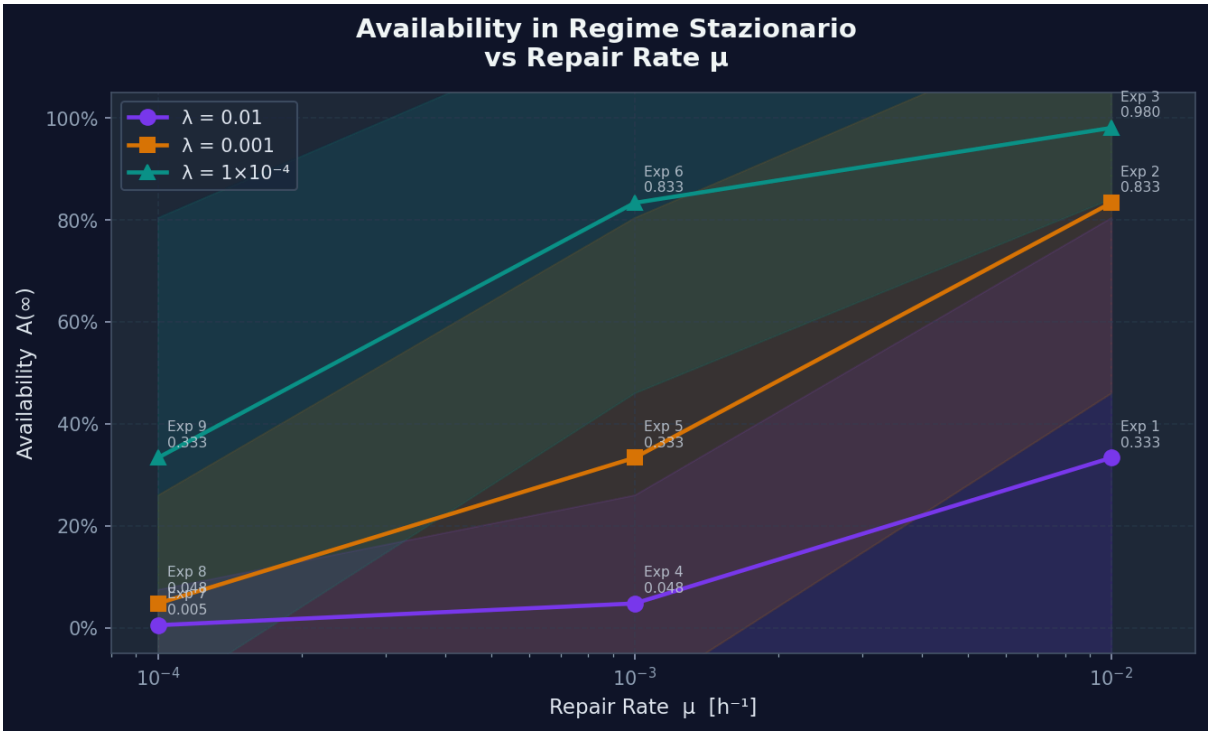
In the R1-O1-W1 scenario, this metric quantifies the fraction of the total time in which at least one functional Source–Sink path exists, implicitly accounting for repair and reactivation policies. The closer the value is to 1, the greater the degree of resilience and global operability. The results of the 9 experiments cover the entire spectrum: from $A = 0.49\%$ (Exp 7) to $A = 98.04\%$ (Exp 3).

Observation 1 — Availability vs Failure Rate



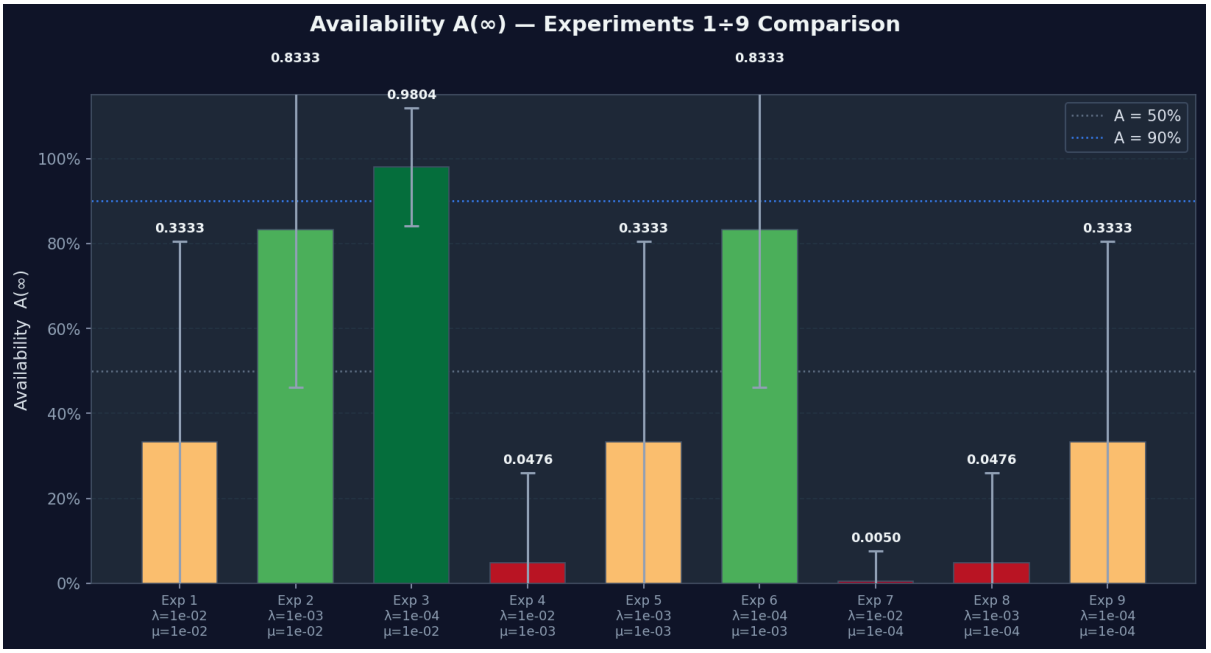
The Availability increases monotonically as λ decreases, regardless of the value of μ. For the same repair rate, lowering λ by an order of magnitude produces availability gains even greater than 90%. The curve with $\mu = 10^{-4}$ (slow repair) shows the maximum sensitivity: the system is condemned to $A < 5\%$ with $\lambda = 10^{-2}$, but reaches $A \approx 33\%$ when $\lambda = \mu$. The confidence band ($\pm 1\sigma$) shows greater variance and stochastic uncertainty in the intermediate/transitional configurations.

Observation 2 — Availability vs Repair Rate (μ)



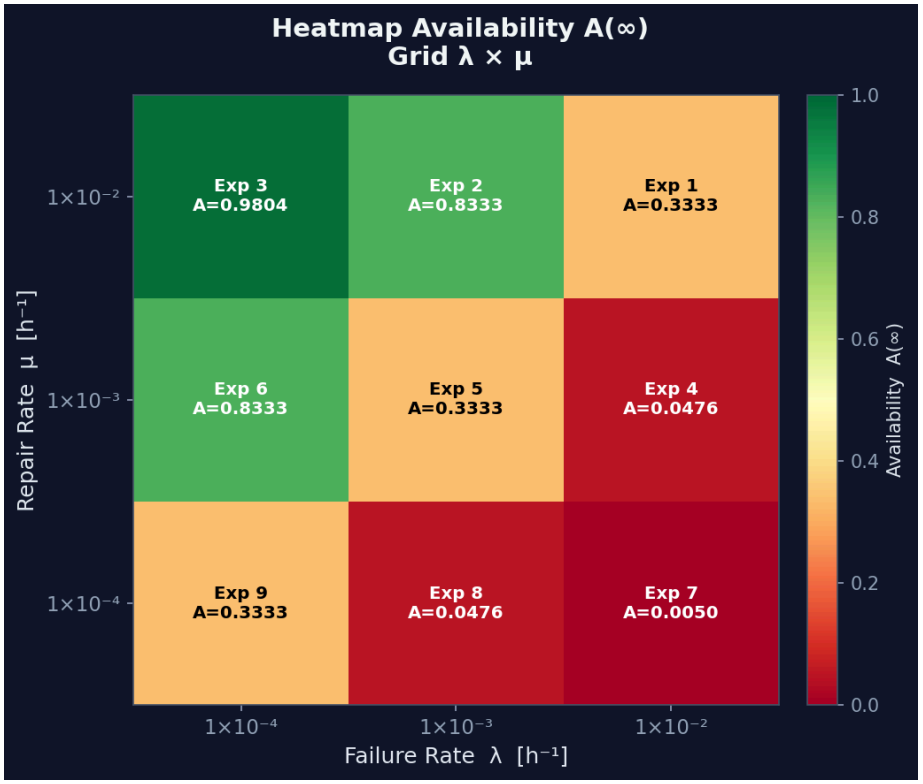
Conversely, Availability increases with μ. The curve at $\lambda = 10^{-2}$ remains consistently depressed even with fast repairs: Exp 1 ($\lambda = \mu = 10^{-2}$) stabilizes at $A = 33.3\%$, a sign that the run-to-failure policy with the SPOF (Unit A) imposes a structural limit on maximum availability. Conversely, with $\lambda = 10^{-4}$, fast repairs ($\mu = 10^{-2}$) lead to $A > 98\%$ (Exp 3): the system is almost always operational.

Observation 3 — Global comparison between all experiments



The bar chart summarizes the dispersion among the 9 experiments. The diagonal symmetry emerges: experiments with $\lambda = \mu$ (Exp 1, 5, 9) all share $A = 33.3\%$ independently of the absolute magnitude of the rates, which is direct evidence of the exclusive dependence on $\rho = \lambda/\mu$. The error bars ($\pm 1\sigma$) show maximum variance in the configurations with $A \approx 33\%$, which corresponds to the maximum of the balanced Bernoulli distribution. The extremes (Exp 3 with $A \approx 98\%$ and Exp 7 with $A < 0.5\%$) delimit the absolute limits of the system.

Observation 4 — Grid Heatmap $\lambda \times \mu$



The heatmap offers the most immediate reading of the experimental space. Green concentrates in the top-left corner (λ low, μ high), red in the bottom-right corner (λ high, μ low). The diagonal tends towards yellow ($A \approx 33\%$), confirming the invariance when $\lambda = \mu$. This "diagonal gradient" pattern is the visual signature of the equation $A(\infty) = \mu/(\lambda + \mu)$.

4.2 Transient Reliability

The Reliability R(t) expresses the probability that the system has not suffered any critical failure in the interval $[0, t]$, starting from an intact state at $t = 0$. For the exponential model, it holds:

$$R(t) = e^{-\lambda_{eff} \cdot t}$$

Experimental Design: unlike Availability, Reliability depends exclusively on λ. To make this structure explicit, the graphs are organized by fixing μ to a constant value and observing how R(t) varies as λ changes. The three values of μ are then compared to experimentally demonstrate the invariance.

Observation 1 — $R(t)$ with μ fixed for each

Three graphs are shown, one for each fixed value of μ, each with the three curves corresponding to the three values of λ.







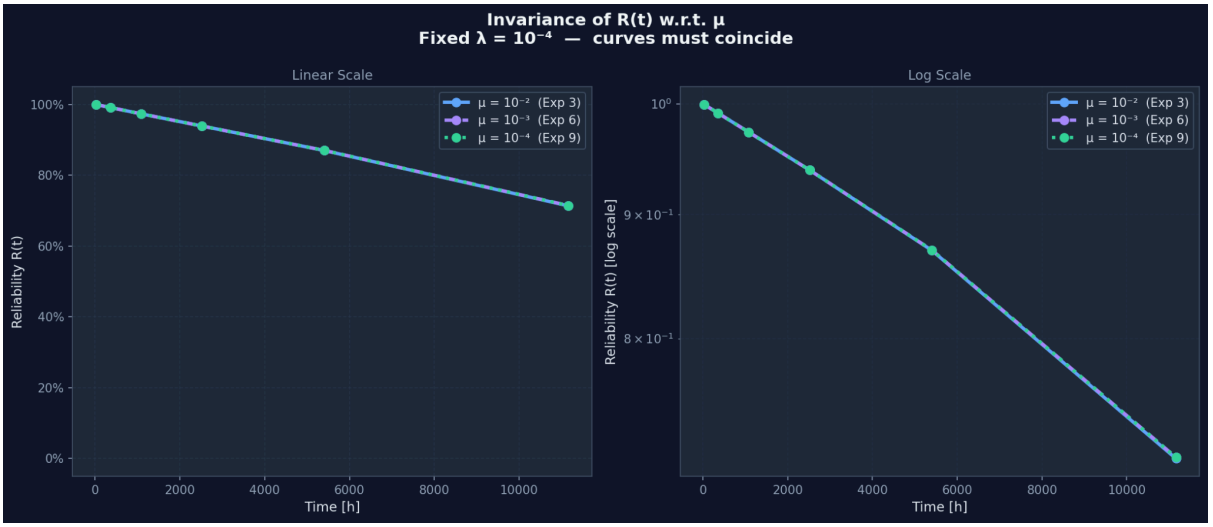
In each graph, the same pattern is observed:

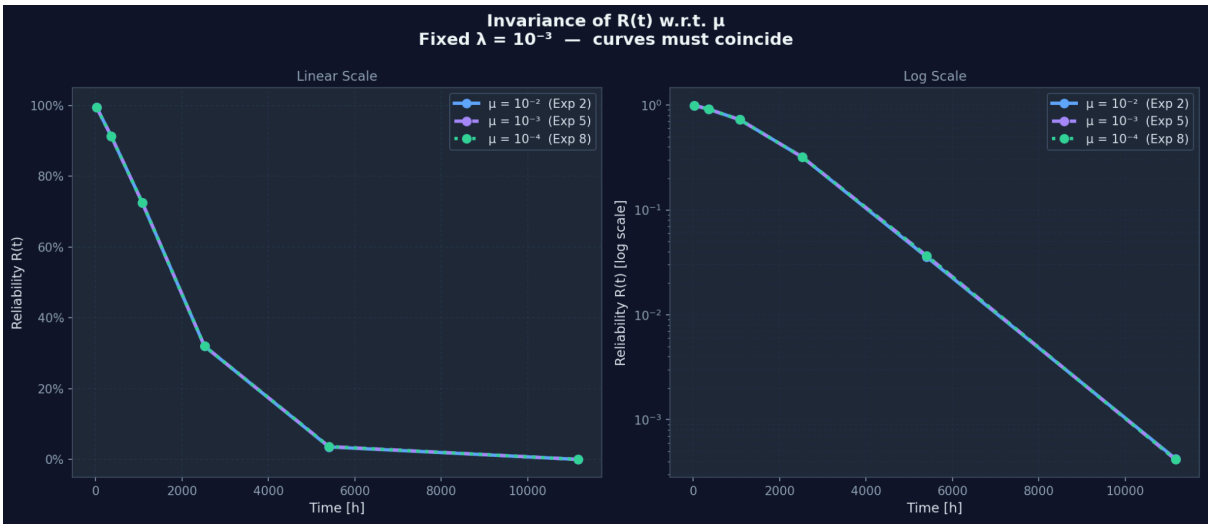
- $\lambda = 10^{-2}$ (Exp 1, 4, 7): rapid collapse. At t = 360 h, Reliability is at 25%; at t = 1080 h, it is practically null. MTTF = 100 hours.
- $\lambda = 10^{-3}$ (Exp 2, 5, 8): gradual degradation. It exceeds 90% up to ~100 hours, drops below 10% around 5000 hours. MTTF = 1000 hours.
- $\lambda = 10^{-4}$ (Exp 3, 6, 9): slow degradation. At t = 11160 h (over 1 year), Reliability remains above 74%. MTTF = 10,000 hours.

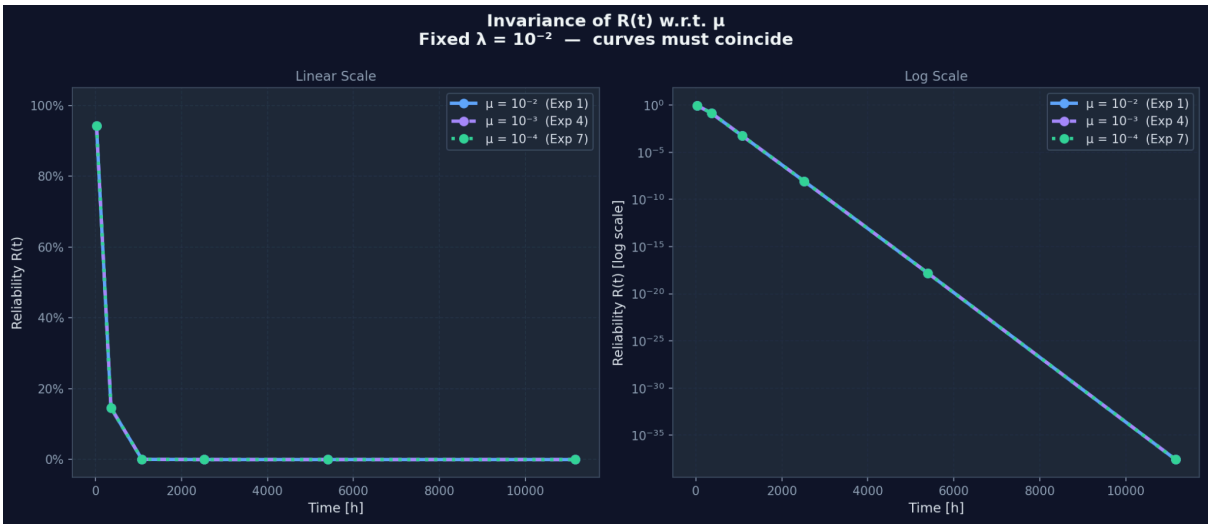
The three graphs are visually identical: first confirmation of the invariance with respect to μ. In fact, due to system requirement R1, repairs are carried out only when the system has failed, thus making the repair frequency invariant for reliability.

Observation 2 — Direct demonstration of the invariance with respect to μ

For each value of λ, the three curves corresponding to the three different μ are superimposed. NOTE: The graphs on the right represent the logarithmic curves. If the hypothesis is correct, the curves must perfectly coincide:

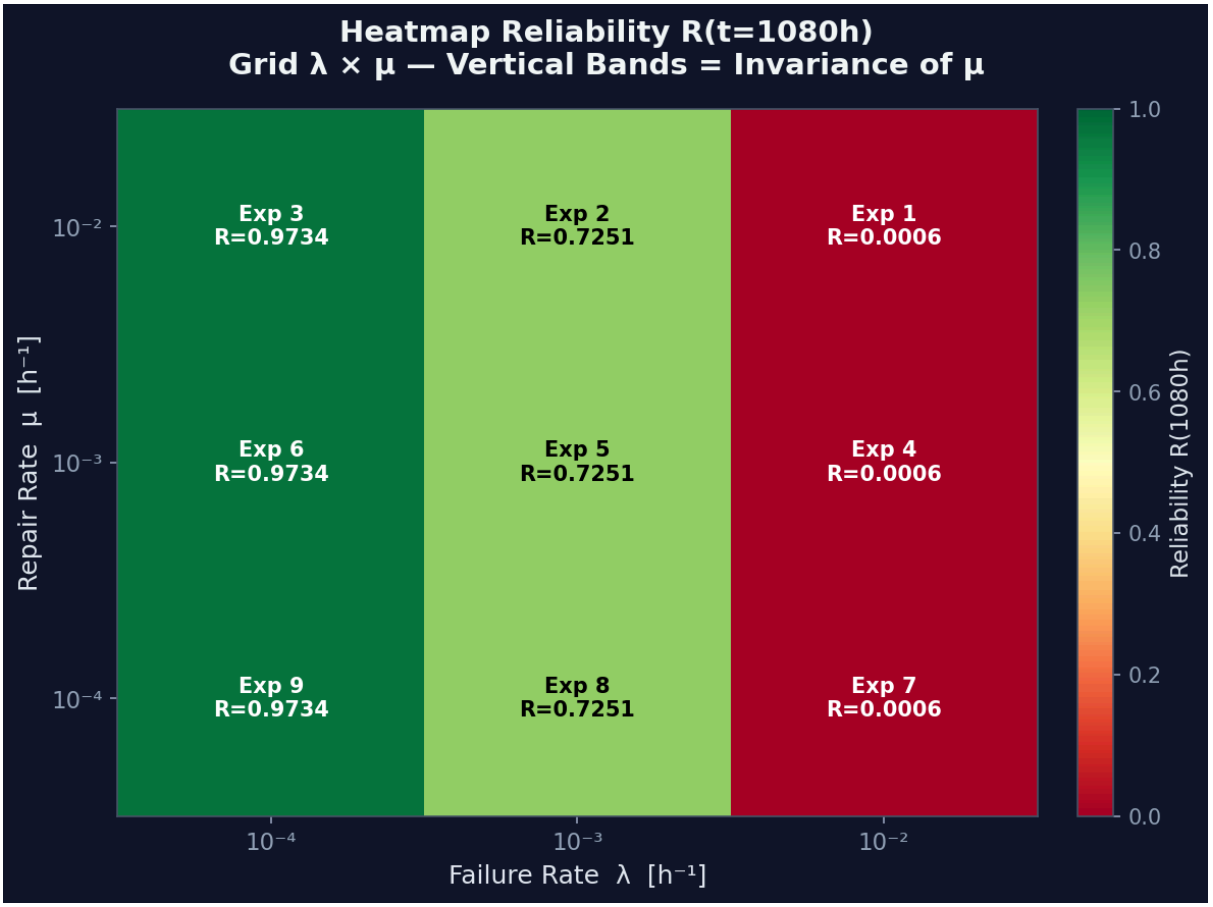

Invariance of R(t) w.r.t. μ
Fixed λ = 10⁻⁴ — curves must coincide
Linear Scale
Log Scale
μ = 10⁻² (Exp 3)
μ = 10⁻³ (Exp 6)
μ = 10⁻⁴ (Exp 9)
Reliability R(t)
Reliability R(t) [log scale]
Time [h]


Invariance of R(t) w.r.t. μ
Fixed λ = 10⁻³ — curves must coincide
Linear Scale
Log Scale
μ = 10⁻² (Exp 2)
μ = 10⁻³ (Exp 5)
μ = 10⁻⁴ (Exp 8)
Reliability R(t)
Reliability R(t) [log scale]
Time [h]


Invariance of R(t) w.r.t. μ
Fixed λ = 10⁻² — curves must coincide
Linear Scale
Log Scale
μ = 10⁻² (Exp 1)
μ = 10⁻³ (Exp 4)
μ = 10⁻⁴ (Exp 7)
Reliability R(t)
Reliability R(t) [log scale]
Time [h]

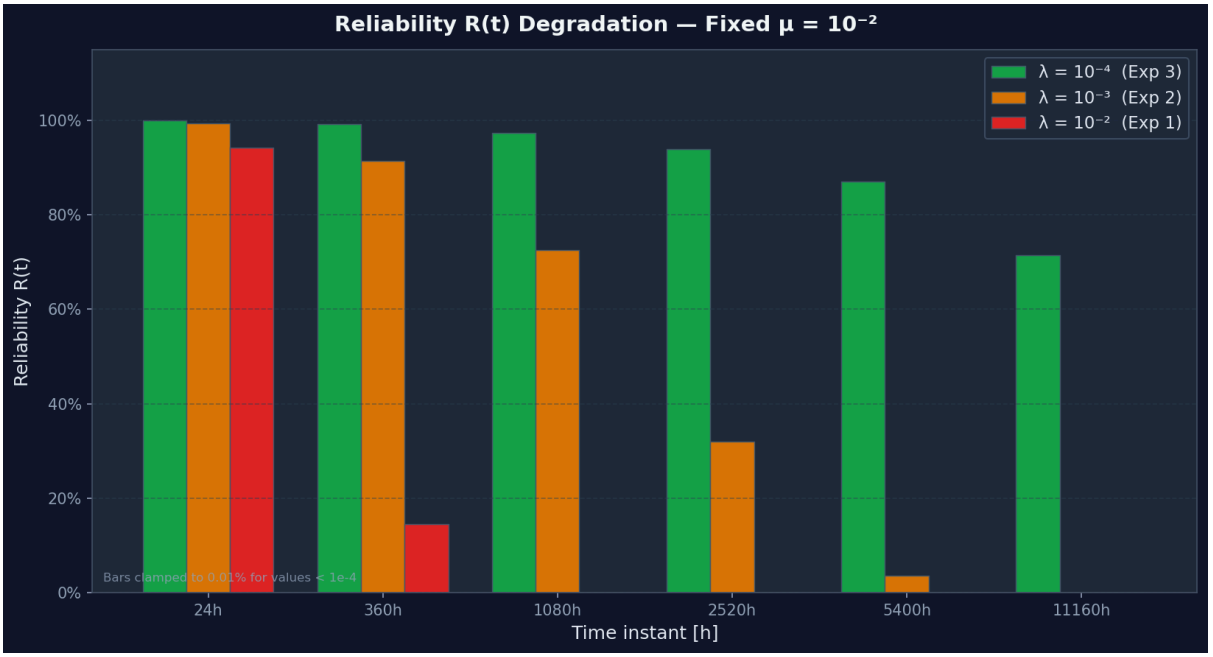
In all three cases, the curves overlap exactly. This is the most important experimental result of the section: the repair speed has no effect on Reliability.

Observation 3 — Heatmap Reliability at t = 1080 h



The heatmap at $t = 1080$ h (45 days) visually encodes the vertical banding structure: all cells in the same column (same x) show identical values, regardless of the row (y). This sharply contrasts with the Availability heatmap (diagonal gradient), graphically confirming the exclusive role of x in determining $R(t)$.

Observation 4 — Bar Chart: degradation at multiple time instants ($\mu = 10^{-2}$ fixed)



The bar chart sets $\lambda_s = 10^{-2}$ (Exp 1, 2, 3) and compares the degradation of R(t) at the six sampled time instants for the three classes of λ_d:

- The Reliability for $\lambda_d = 10^{-2}$ (red) collapses within the first 360 hours;
- The Reliability for $\lambda_d = 10^{-4}$ (green) maintains high values over the entire time horizon.

The bars are truncated at the minimum 10^{-4} for readability; the real values for t > 2520 h with $\lambda_d = 10^{-2}$ are numerically close to zero. Given the invariance of λ_s for Reliability, the graph has the same structure for any value of λ_s.

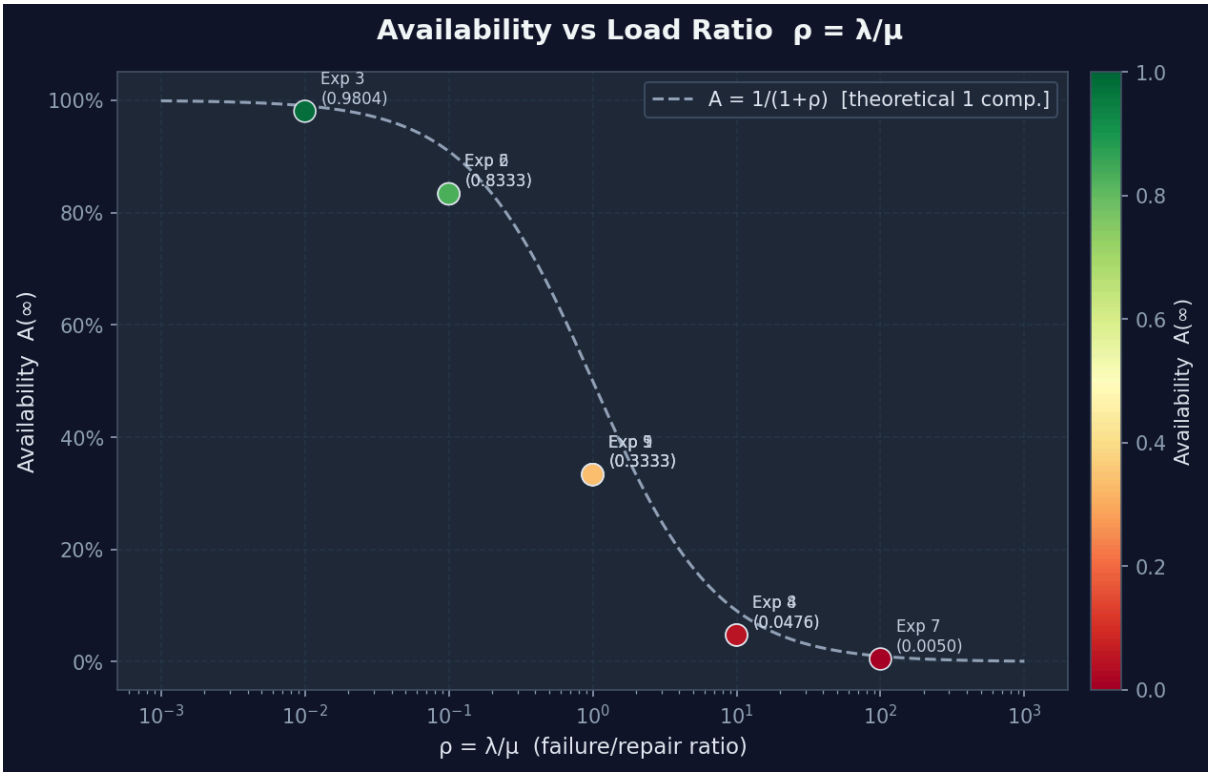
4.3 Load Ratio (ρ)

In addition to the standard metrics, we have introduced the dimensionless ratio ρ, indirectly derived from the experimental parameters:

$$\rho = \frac{\lambda}{\mu}$$

This synthetic indicator measures the maintenance effort the system is subjected to. When $\rho \ll 1$ repairs dominate failures and Availability is high; when $\rho \gg 1$ the system is structurally overwhelmed by failures. Its main utility is to collapse the dependence on two parameters (λ, μ) into a single scalar quantity, making experiments with different absolute magnitudes comparable.

Observation — Availability vs ρ and theoretical curve



The graph superimposes the 9 experimental points on the theoretical curve $A = 1/(1+\rho)$. The correspondence is almost perfect: it confirms that the CTMC model with policies R1-O1-W1 converges in steady state exactly to the system's analytical solution. Experiments with the same ρ (e.g., Exp 1, 5, 9 with $\rho = 1$) overlap on the same point of the curve, independently of the absolute magnitude of λ and μ. The logarithmic scale on the X-axis allows visualizing the entire range from $\rho = 10^{-2}$ (optimally maintained system) to $\rho = 10^{2}$ (neglected system). The coloring of the points (green = high Availability, red = low Availability) reinforces the immediate reading of the curve.